

1. What is a Function? State its Types and Explain the Syntax to Define a Function.

A **function** is a block of reusable code that performs a specific task. Functions help make programs modular, organized, and easy to maintain.

Types of Functions

a) Built-in Functions

Functions that are already available in Python.

Examples:

```
print()
len()
sum()
max()
```

b) User-Defined Functions

Functions created by the programmer using the `def` keyword.

Syntax of a Function

```
def function_name(parameters):
    # function body
    return value
```

Example

```
def add(a, b):
    return a + b
```

```
result = add(10, 20)
print(result)
```

Output:

```
30
```

Components of a Function

`def` → Keyword used to define a function.

`function_name` → Name of the function.

`parameters` → Input values passed to the function.

`return` → Sends a value back to the caller.

2. What is a Lambda Function?

A lambda function is a small anonymous function that can have any number of arguments but only one expression.

Syntax

```
lambda arguments : expression
```

Example

```
square = lambda x: x * x
```

```
print(square(5))
```

Output:

```
25
```

Another Example

```
add = lambda a, b: a + b
```

```
print(add(10, 20))
```

Output:

```
30
```

Advantages of Lambda Functions

Short and concise.

Useful for simple operations.

Commonly used with functions like `map()`, `filter()`, and `sorted()`.

Example with `filter()`

```
numbers = [1, 2, 3, 4, 5, 6]
```

```
even = list(filter(lambda x: x % 2 == 0, numbers))
```

```
print(even)
```

Output:

```
[2, 4, 6]
```